



Endogenous fluctuations with procyclical R&D

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ABSTRACT

The literature on endogenous growth cycles predicts the countercyclical allocation of resources to R&D. However, this prediction is not supported by empirical studies. This study considers the R&D-based growth model with endogenous fluctuations introducing population growth and a negative externality that affects the productivity of R&D. We show that this simple modification makes R&D investment procyclical along sustained business cycles using both an overlapping generation framework and an infinitely-lived agent framework.

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1. Introduction

This study examines how R&D activity varies over the business cycle using the framework of the R&D-based growth models. Studies of the interaction between R&D and endogenous fluctuations were pioneered by Judd (1985) and Deneckere and Judd (1992). By applying a bifurcation theorem, they found fluctuating equilibrium paths in the variety-expansion model without capital accumulation. However, in the above mentioned models, a sustained R&D effort did not contribute to long-run growth because it was canceled out by the obsolescence and dilution of knowledge. Matsuyama (1999, 2001) has modified the model in Deneckere and Judd (1992) by introducing capital accumulation and Romer's (1990) idea of endogenous growth,¹ and investigated endogenous fluctuations with sustained long-run growth.² One of the main predictions of these theoretical models was the countercyclical behavior of R&D over sustained growth cycles; that is, with low growth, resource allocation to R&D is high. In addition, this countercyclicity of R&D made productivity improvements countercyclical.³

Francois and Lloyd-Ellis (2003) have studied the endogenous growth model with endogenous fluctuations, which was based on the quality-ladder framework of Grossman and Helpman (1991) and the theory of the implementation cycles of Shleifer (1986). In their model, productivity improvements were procyclical; however R&D expenditures were still countercyclical. Other studies, including Wälde (2002), Bental and Peled (1996), and Francois and Lloyd-Ellis (2008), predicted the countercyclical behavior of R&D.⁴

The prediction that R&D expenditures are countercyclical is difficult to justify from empirical studies. Wälde and Woitek (2004) have studied the cyclical properties of R&D in G7 countries using annual data from 1973 to 2000. They found that aggregate R&D expenditures tended to be procyclical and argued that the prediction of Matsuyama (1999, 2001) was counterfactual. Fatás (2000) and Comin and Gertler (2006) also have found a highly procyclical tendency of R&D expenditures using U.S. data. In particular, Comin and Gertler (2006) focused on longer-term oscillations than conventional business cycles. They termed these oscillations the “medium-term cycle” that includes frequencies between 6 months and 50 years. In this respect, there is a close relationship between their empirical study and our theoretical analysis. Geroski and Walters (1995) argued that their analysis of the U.K. data revealed that productivity improvements were also procyclical. Barlevy (2007), using data from both the National Science Foundation (NSF) and Standard & Poor's Compustat database of publicly traded

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¹ In Matsuyama (2001), capital accumulation is derived from intertemporal optimization of the infinitely lived agents. Matsuyama (1999) did not present intertemporal optimization explicitly.

² More precise studies of the dynamics of Matsuyama's model were presented by Mitra (2001), Mukherji (2005), Gardini et al. (2008), and Yano et al. (2011).

³ In business cycles, when any economic quantity is positively correlated with the business condition of the economy, it is said to be procyclical. Countercyclical is the opposite of procyclical. In this study, the growth rate of real GDP is used as a procyclical economic indicator.

⁴ Francois and Lloyd-Ellis (2008) did not interpret the activity that was a source of productivity improvements as R&D, but as an “entrepreneurial search.” However, its process was formally identical to the R&D process in the earlier models of Grossman and Helpman (1991) and Francois and Lloyd-Ellis (2003).

companies, found a positive correlation between the growth rate of R&D at the industry level as well as the aggregate level.

The main purpose of this paper is to include the procyclical behavior of R&D into the endogenous fluctuation model. We modify the variety-expansion model in Matsuyama (1999, 2001), introducing population growth and a negative externality that affects the productivity of R&D. We assume that finding new knowledge becomes more difficult as economies become technologically more advanced, as in the semi-endogenous growth model in Jones (1995) and Segerstrom (1998).⁵ This assumption has been first proposed to eliminate the scale effect, which is a serious counterfactual prediction in the first-generation R&D-based endogenous growth models, such that an economy with a large population grows faster.⁶

Relevant related literature includes Wälde (2005), Francois and Lloyd-Ellis (2009), Comin and Gertler (2006), and Barlevy (2007). Francois and Lloyd-Ellis (2009) have studied the endogenous business cycle model based on their previous work (Francois and Lloyd-Ellis, 2003). They decomposed the innovation process into three distinct stages: R&D, commercialization, and innovation. Their model illustrated the procyclical movement of R&D, and they determined that the countercyclical movement of commercialization played a central role in this new result. Furthermore, they showed that the total expenditure for innovation, defined as the sum of expenditures for R&D and commercialization, moved procyclically. Wälde (2005) also illustrated procyclical R&D behavior by using a quality-ladder framework with capital accumulation. The Francois and Lloyd-Ellis (2009) and Wälde (2005) models are similar to ours in that they assumed a negative externality of knowledge accumulation and derive non-scale growth with endogenous fluctuations. On the other hand, Comin and Gertler (2006) and Barlevy (2007) have discussed the cyclicity of R&D over the business cycles that were caused by exogenous shocks. The former was based on a variety-expansion framework and used similar approach to Francois and Lloyd-Ellis (2009), i.e., decomposing the innovation process.⁷ The latter, using a quality-ladder framework, showed that the equilibrium R&D was procyclical in a decentralized market. However, optimal R&D was found to be countercyclical by a central planner's problem.⁸

As the aforementioned studies illustrate, the theoretical explanation of the procyclicity of R&D is one of the most controversial topics in the studies of R&D and business cycles. This study achieves the procyclical R&D behavior under an assumption that is simpler than those of Francois and Lloyd-Ellis (2009) and Wälde (2005). In addition, it does not require exogenous shocks, unlike the Comin and Gertler (2006) and Barlevy (2007) models.

The rest of this paper is organized as follows. Section 2 sets up the model used in our theoretical investigation and derives the law of motion that characterizes the equilibrium path of the economy. Section 3 examines the dynamic properties of the model and illustrates that the equilibrium path fluctuates endogenously. Section 4 focuses on period 2 cycles and studies the cyclicity of R&D investment. Section 5 provides conclusions.

2. The model

Our model considers the dynamic model based on Matsuyama (1999). Time is discrete and indexed by $t = 0, 1, 2, \dots$. There is a single final good taken as a numeraire that is produced using intermediate goods and labor. It can be consumed or invested. A new variety of intermediate goods is invented by allocating capital for R&D activities.

⁵ Jones (1995) called such an externality the fishing-out effect.

⁶ Jones (1995) was also the study based on the variety-expansion model in Romer (1990). However, its balanced growth path (BGP) has a saddle property and no endogenous fluctuation occurs as proven by Arnold (2006). Note that in order to examine the dynamics analytically, Arnold (2006) assumed constant returns to labor in R&D, which was not assumed in Jones' original model.

⁷ They introduced the stage of "adoption" instead of commercialization.

⁸ For other recent work on R&D procyclicity, see Nuño (2011).

Inventors enjoy a one-period monopoly by patent protection. The available intermediate goods are produced by multiple intermediate firms using capital. Finally, we assume two-period-lived overlapping generation (OLG) households, who inelastically supply labor when young.

2.1. Final goods

We assume that perfect competition prevails in the final goods market. The production function is given by

$$Y_t = AL_t^{1-\alpha} \int_0^{N_t} x_t(z)^\alpha dz, \quad 0 < \alpha < 1, \quad A > 0, \quad (2.1)$$

where Y_t is the final output, L_t is the inelastically supplied labor, $x_t(z)$ is the amount of the intermediate good indexed by z , and $1/(1-\alpha)$ denotes the elasticity of substitution between all pairs of intermediate goods. N_t is the number of available intermediate goods in period t that represents the technology level of the economy.

Profit maximization yields $w_t = (1-\alpha)Y_t/L_t$ and the inverse demand function for each intermediate good $z \in (0, N_t]$ as $p_t(z) = \alpha AL_t^{1-\alpha} x_t(z)^{-(1-\alpha)}$, where w_t is the real wage rate and $p_t(z)$ is the price of the intermediate good z .

2.2. Intermediate goods

Each intermediate good is produced by using one unit of capital. Let r_t denote the price of capital. Because of limited patent protection, the "old" intermediate goods, $[0, N_{t-1})$, are supplied competitively. Hence, the price is equal to the marginal cost, $p_t(z) = r_t$, for $z \in (0, N_{t-1}]$. However, the "new" intermediate goods invented in period $t-1$, $(N_{t-1}, N_t]$, are supplied monopolistically and sold at the monopoly price, $p_t(z) = r_t/\alpha$, for $z \in (N_{t-1}, N_t]$. All intermediate goods enter symmetrically into the production of the final good, i.e., $x_t(z) = x_{ct}$ for $z \in [0, N_{t-1}]$ and $x_t(z) = x_{mt}$ for $z \in (N_{t-1}, N_t]$. We can easily illustrate that $x_{mt} = \alpha^{-1/\alpha} x_{ct}$ holds and the maximized monopoly profits are $\Pi_t(z) = \Pi_t \equiv \frac{1-\alpha}{\alpha} x_{mt} r_t$ for $z \in (N_{t-1}, N_t]$.

2.3. R&D

The number of intermediate goods N expands according to the following equation:⁹

$$N_t - N_{t-1} = \eta \frac{R_t}{N_{t-1}^\phi}, \quad N_0 > 0, \quad \phi > 0, \quad \eta > 0,$$

where R_t is the amount of the capital allocated to R&D. Following the formation adopted in Jones (1995), we assume that the past discoveries make inventing a new machine more difficult. This external effect is captured by ϕ .

Each inventor enjoys a one-period monopoly and earns profits Π_t . Therefore, in equilibrium, the following free-entry condition must be satisfied:

$$\Pi_t \leq \eta^{-1} N_{t-1}^\phi r_t, \quad \text{with an equality whenever } N_t > N_{t-1}.$$

The break-even point of x_{mt} is given by $\bar{x}_{mt} \equiv \frac{\alpha}{1-\alpha} \eta^{-1} N_{t-1}^\phi$. It becomes larger for a large value of ϕ , since R&D becomes costlier for any given N_{t-1} and L_t .

Finally, clearing the capital market requires $K_t = R_t + (N_t - N_{t-1})x_{mt} + N_{t-1}x_{ct}$, where K_t is the amount of capital accumulated in period $t-1$ and available in period t . The available capital is utilized by R&D, producing monopolistic intermediate goods, and producing competitive intermediate goods.

⁹ This specification is based on Rivera-Batiz and Romer's (1991) "lab equipment model."

2.4. Consumers

Each consumer lives for two periods. When young, he/she supplies one unit of labor and earns wage w_t , which is divided into savings and consumption. When old, he/she only consumes using his/her savings. Let c_{1t} and c_{2t+1} denote the consumption in periods t and $t+1$, respectively, of the consumers born in period t . Each consumer chooses c_{1t} and c_{2t+1} that maximizes their utility, $U_t = (1-s)\log c_{1t} + s\log c_{2t+1}$, where $s \in (0,1)$, subject to the budget constraint, $c_{2t+1} = (w_t - c_{1t})r_{t+1}$.

The solution to this simple maximization problem is characterized by the linear saving function, $K_{t+1} = sw_t L_t = s(1-\alpha)Y_t$, where L_t represents the number of consumers born in period t , which increases at the exogenous rate $n > 0$, i.e., $L_t = (1+n)L_{t-1}$.

2.5. Equilibrium

We obtain the following equation in a very similar way to Matsuyama (2001, Eq. (9)):

$$\frac{N_t - N_{t-1}}{N_{t-1}} = \mu(k_{t-1}) \equiv \max\left\{0, \alpha^{-\frac{\alpha}{1-\alpha}}(k_{t-1} - 1)\right\}, \quad (2.2)$$

where we define $k_{t-1} \equiv \alpha^{\frac{\alpha}{1-\alpha}}(1-\alpha)\eta K_t / N_{t-1}^{1+\phi}$. If $k_{t-1} > 1$ holds, i.e., the economy has a sufficient stock of capital relative to its technological level, the positive amount of capital is allocated for R&D and $N_t > N_{t-1}$ holds. In contrast, if $k_{t-1} \leq 1$, neither R&D occurs nor technological progress arises.

Substituting Eq. (2.2) into Eq. (2.1) illustrates that the total output is equal to

$$Y_t = \frac{AL_t^{1-\alpha} N_{t-1}^{1+\alpha\phi}}{\alpha^{\frac{\alpha}{1-\alpha}}(1-\alpha)\eta} \psi(k_{t-1}), \quad \psi(k_{t-1}) \equiv \begin{cases} k_{t-1}^\alpha, & \text{if } k_{t-1} \leq 1, \\ k_{t-1}, & \text{if } k_{t-1} > 1. \end{cases} \quad (2.3)$$

To describe the equilibrium path of this economy, we define the new variable $\ell_{t-1} \equiv [s(1-\alpha)A]^{\frac{1-\alpha}{1+\phi}} \alpha^{\frac{\alpha}{1-\alpha}}(1-\alpha)\eta L_t / N_{t-1}^\phi$. Summarizing the above equations yields the following two-dimensional dynamical system:

$$\begin{aligned} \ell_t = f^\ell(k_{t-1}, \ell_{t-1}) &= \begin{cases} (1+n)\ell_{t-1}, & \text{if } k_{t-1} \leq 1, \\ \frac{(1+n)\ell_{t-1}}{[1+\alpha^{-\frac{\alpha}{1-\alpha}}(k_{t-1}-1)]^\phi}, & \text{if } k_{t-1} > 1, \end{cases} \\ k_t = f^k(k_{t-1}, \ell_{t-1}) &= \begin{cases} \ell_{t-1}^{1-\alpha} k_{t-1}^\alpha, & \text{if } k_{t-1} \leq 1, \\ \frac{\ell_{t-1}^{1-\alpha} k_{t-1}}{[1+\alpha^{-\frac{\alpha}{1-\alpha}}(k_{t-1}-1)]^{1+\phi}}, & \text{if } k_{t-1} > 1. \end{cases} \end{aligned} \quad (2.4)$$

If the initial values of k_0 and ℓ_0 are given, the law of motion, Eq. (2.4), characterizes the equilibrium path $\{k_t, \ell_t\}_0^\infty$, whose properties depend on parameter values, α , n , and ϕ .

3. Dynamics

The law of motion, Eq. (2.4), has a unique positive fixed point (k^*, ℓ^*) , where

$$k^* = 1 + \alpha^{\frac{\alpha}{1-\alpha}} \left[(1+n)^{\frac{1}{\phi}} - 1 \right] > 1, \quad \ell^* = (1+n)^{\frac{1+\phi}{\phi(1-\alpha)}}. \quad (3.1)$$

In the long run, $k \leq 1$ is unsustainable by an exogenous population growth. Therefore, (k^*, ℓ^*) is a unique non-trivial fixed point of the dynamical system (Eq. (2.4)).¹⁰ At this fixed point, K , Y (or per capita output

¹⁰ Substituting $\ell_t = \ell_{t-1} > 0$ into $\ell_t = f^\ell(k_{t-1}, \ell_{t-1})$, and solving for k_{t-1} , we obtain k^* uniquely. ℓ^* uniquely exists by corresponding with a unique k^* .

$y \equiv Y_t / (L_{t-1} + L_t)$), and N grow at constant rates. That is, the economy achieves balanced growth. The balanced growth rate of per capita output is derived as $g_y^* = (1+n)^{\frac{1}{\phi}} - 1$, which is independent of population L .

3.1. Stability

The two-dimensional system, Eq. (2.4), has two predetermined variables, k and ℓ . If the fixed point is a sink, it is locally stable.

Proposition 1. *There is a unique bifurcation point of ϕ , ϕ_b , that satisfies $B(\phi_b) - \Lambda(\phi_b) = 0$. The fixed point (k^*, ℓ^*) is a sink for $\phi < \phi_b$, whereas it is a saddle point for $\phi > \phi_b$, where $B(\phi)$ and $\Lambda(\phi)$ are defined as follows:*

$$B(\phi) \equiv \frac{2 - \phi(1+\alpha)}{2 + \phi(1+\alpha)}, \quad \Lambda(\phi) \equiv \frac{\alpha^{\frac{\alpha}{1-\alpha}} - 1}{(1+n)^{\frac{1}{\phi}}}.$$

Furthermore, in the sufficiently small neighborhood of (k^*, ℓ^*) , the system (Eq. (2.4)) has a periodic orbit of period-2 on one side of the bifurcation point ϕ_b .

Proof. See Appendix A.

Proposition 1 argues that the unique fixed point loses its stability for sufficiently large ϕ . Moreover, a flip bifurcation (period-doubling bifurcation) occurs by slightly changing a bifurcation parameter ϕ . If this bifurcation is supercritical, there are stable period 2 cycles for $\phi > \phi_b$ in the neighborhood of ϕ_b . Conversely, if the bifurcation is subcritical, period 2 cycles with a saddle property for $\phi < \phi_b$.

4. Period 2 cycles

The existence of the period 2 cycles is verified by the following method.

Proposition 2. *If and only if $\phi \geq \bar{\phi}_1$, the system, Eq. (2.4), has a pair of the fixed points of period 2, (k^H, ℓ^H) and (k^L, ℓ^L) , such that $k^L \leq 1 < k^H$, where $\bar{\phi}_1$ is defined as follows:*

$$\bar{\phi}_1 \equiv \frac{\log(1+n)}{\log \chi}, \quad \text{where } \chi \equiv \frac{(1+n)^{\frac{1+\alpha}{2}} + \sqrt{(1+n)^{1+\alpha} - 4\alpha^{\frac{\alpha}{1-\alpha}}(1-\alpha^{\frac{\alpha}{1-\alpha}})}}{2\alpha^{\frac{\alpha}{1-\alpha}}}.$$

Proof. See Appendix B.

When the parameters satisfy the conditions of Proposition 2, the system, Eq. (2.4), has the period 2 cycles moving back and forth between the two phases, as shown in Deneckere and Judd (1992) and Matsuyama (1999). In one phase, capital is allocated to R&D and new intermediate goods are invented. In the other phase, all capital is allocated to the intermediate goods sector and no invention occurs. We shall refer to each phase as the *R&D phase* and the *no R&D phase*, respectively. The average growth rates of per capita output over the cycles are given by $g_y^{\text{cycle}} = (1+n)^{\frac{1}{\phi}} - 1$, which is equal to the growth rate along its balanced growth path (BGP), g^* .

Our main purpose is to clarify whether R&D investment is procyclical or countercyclical over business cycles. Let g_X^1 denote the growth rate of the variable X in the R&D phase. Similarly, g_X^0 denotes the growth rate of X in the no R&D phase.

Proposition 3. (a) *If $\alpha > 1/2$, a threshold $\bar{\phi}_2$ exists such that $g_y^1 > g_y^0$ holds for $\phi \in [\bar{\phi}_1, \bar{\phi}_2)$, while $g_y^1 < g_y^0$ holds for $\phi > \bar{\phi}_2$ where*

$$\bar{\phi}_2 \equiv \frac{\log(1+n)}{\log(1-\alpha^{\frac{\alpha}{1-\alpha}}) - \log \alpha^{\frac{\alpha}{1-\alpha}}} > \bar{\phi}_1.$$

(b) *If $\alpha \leq 1/2$, $g_y^1 > g_y^0$ holds for any $\phi \geq \bar{\phi}_1$.*

Proof. See Appendix C.

$g_y^1 > g_y^0$ means that the R&D phase achieves faster growth than the no R&D phase; that is, R&D investment is procyclical. In contrast, when $g_y^1 < g_y^0$, R&D investment is high with low growth.

The results of Propositions 2 and 3 appear in Fig. 1 with α on the horizontal axis, ϕ on the vertical axis, and two downward curves. The region above the graph of $\bar{\phi}_1$ corresponds to the set of α and ϕ for which period 2 cycle, described in Proposition 2, exist. Furthermore, that region is separated into two regions by the graph of $\bar{\phi}_2$. The lower region corresponds to the set of parameters associated with procyclical R&D investment, while the upper region corresponds to the set of parameters with the countercyclical R&D investment. We can see that when $\alpha \leq 1/2$, in which case the elasticity of substitution between each intermediate good is low or the mark-up of the monopoly price is high, there does not exist a value of ϕ that causes the countercyclical R&D investment. Whereas, when $\alpha > 1/2$, the sign of the inequality between g_y^1 and g_y^0 may change depending on the values of ϕ .

4.1. Examples

We assume that the parameter values are $\alpha = \{1/3, 0.9\}$ and $n = (1.012)^{10} - 1 \approx 0.1267$. The rate of population growth chosen means a population growth rate of 1.2%/year and a patent length of 10 years.

Example 1. If $\alpha = 0.9$, $n = 0.1267$, and $\phi = 0.27$, the fixed point (k^*, ℓ^*) is a saddle point, and the fixed points of period-2 exist. The growth rates of each phase are $(g_y^1, g_y^0) = (0.5552, 0.5558)$. Therefore, R&D investment is countercyclical.

Example 2. If $\alpha = 0.9$, $n = 0.1267$, and $\phi = 0.25$, the fixed point (k^*, ℓ^*) is a saddle point and the fixed points of period-2 exist. The growth rates of each phase are $(g_y^1, g_y^0) = (0.6118, 0.6111)$. Therefore, R&D investment is procyclical.

The value of the parameter, $\alpha = 0.9$, corresponds to the mark-up of the monopoly price, $1/\alpha \approx 1.1111$. For example, Rotemberg and Woodford (1995) estimated as 1.115. $\alpha = 0.9$ is consistent with their estimation. The threshold values of ϕ are $\phi_b = 0.1572$, $\bar{\phi}_1 = 0.1007$, and $\bar{\phi}_2 = 0.2604$.

Example 3. If $\alpha = 1/3$, $n = 0.1267$, and $\phi = 0.61$, the fixed point (k^*, ℓ^*) is a saddle point and the fixed points of period-2 exist. The growth rates of each phase are $(g_y^1, g_y^0) = (0.2458, 0.1869)$. Therefore, R&D investment is procyclical.

In our model, α equals the share of capital, conventionally considered to be around 1/3. For $\alpha = 1/3$, $\phi_b = 0.4298$ and $\bar{\phi}_1 = 0.4280$, R&D moves procyclically regardless of the value of ϕ .

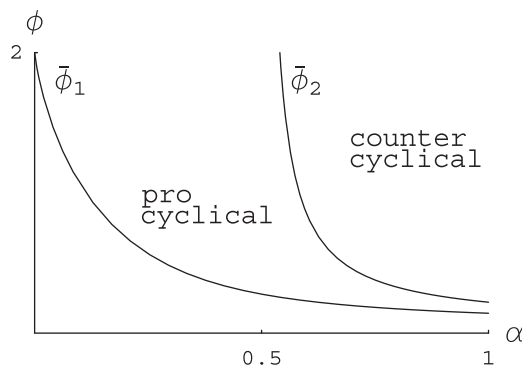


Fig. 1. The cyclicity of R&D.

5. Conclusions

This study has examined the cyclicity of the R&D investment over the business cycles by using the variety-expansion model with limited patent protection. We have illustrated that the unique fixed point loses its stability. In addition, period 2 cycles moving back and forth exist between the R&D and the no R&D phases. Moreover, we examined the possibility and conditions that R&D investment is procyclical over the period 2 cycles and have proven the existence of the parameter set that achieves procyclical R&D, shown in many empirical studies. In our model, countercyclical R&D requires a large capital share and a sufficiently strong external effect. In other cases, R&D investment is procyclical.

We assume exogenous population growth and the negative externality of the stock of knowledge that works in R&D, following the formation of the semi-endogenous growth model.¹¹ In our model, the parameter of this externality plays the central role in the decision of the cyclicity of R&D.

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Appendix A. Proof of Proposition 1

In order to examine local stability, we linearize the system (Eq. (2.4)) around the fixed point (k^*, ℓ^*) .

$$\begin{bmatrix} k_t - k^* \\ \ell_t - \ell^* \end{bmatrix} = J \begin{bmatrix} k_{t-1} - k^* \\ \ell_{t-1} - \ell^* \end{bmatrix}, \quad \text{where } J \equiv \begin{bmatrix} f_1^{k^*} & f_2^{k^*} \\ f_1^{\ell^*} & f_2^{\ell^*} \end{bmatrix}.$$

A stability type of the fixed point depends on the trace ($\text{tr } J$) and the determinant ($\det J$) of the Jacobian matrix. We define seven regions separated by three lines, $\det J = \text{tr } J - 1$, $\det J = -\text{tr } J - 1$, and $\det J = 1$, as shown in Fig. A.2.¹² We also know that if the Jacobian was somehow to move from inside the triangle with sink stability to outside, a bifurcation would occur.

$\det J$ and $\text{tr } J$ are derived as

$$\det J = -\alpha\phi - (1 + \alpha\phi) \frac{\alpha \frac{1-\alpha-1}{1}}{(1+n)\phi}, \quad \text{tr } J = -\phi + 1 - (1 + \phi) \frac{\alpha \frac{1-\alpha-1}{1}}{(1+n)\phi}. \quad (\text{A.1})$$

It is clear that $\det J < 1$ and $\det J > \text{tr } J - 1$, that is, the pair of $\det J$ and $\text{tr } J$ does not belong to the shaded region in Fig. A.2.¹³ In addition, from Eq. (A.1), we obtain the following relationship:

$$\Lambda(\phi) \leq B(\phi) \Leftrightarrow \det J \geq -\text{tr } J - 1.$$

□

¹¹ As a result, even if the economy grows along the fluctuating equilibrium path, the long-run growth is not endogenous and requires positive population growth, as shown in the literature using a similar assumption such as Jones (1995) and Segerstrom (1998). Young (1998), Peretto (1998), and Howitt (1999), indicated this problem and proposed models that have non-scale endogenous growth. A survey of this issue is presented in Jones (1999, 2005). Li (2000, 2002) argued that the predictions of these models depend on the knife-edge assumption and that the semi-endogenous growth prediction is more general.

¹² See Azariadis (1993, Ch.6) for further details.

¹³ Therefore, the possibilities of a saddle-node bifurcation and a Hopf bifurcation can be ruled out.

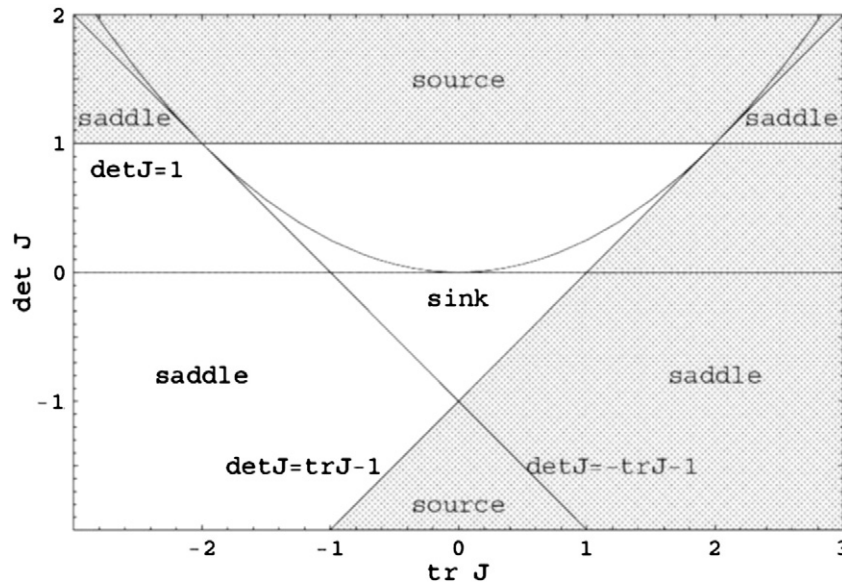


Fig. A.2. Local stability on the plane.

Appendix B. Proof of Proposition 2

Solving $\ell^H = f^{\ell}(f^K(k^H, \ell^H), f^{\ell}(k^H, \ell^H))$ for k^H , we obtain

$$k^H \equiv 1 + \alpha^{\frac{\alpha}{1-\alpha}} \left[(1+n)^{\frac{2}{\phi}} - 1 \right] > 1. \quad (\text{B.1})$$

Furthermore, k^L , ℓ^L , and ℓ^H are

$$k^L \equiv \left[k^H (1+n)^{\frac{1-\alpha}{2} - \frac{1+\phi}{\phi}} \right]^{\frac{2}{1-\alpha}}, \quad (\text{B.2})$$

$\ell^L \equiv (k^H)^{\frac{1}{1-\alpha}} (1+n)^{-\frac{2\alpha}{1-\alpha} \left(\frac{1+\phi}{\phi} - \frac{1-\alpha}{2} \right)}$, and $\ell^H = (1+n)\ell^L$. Certain algebra shows that $k^L \leq 1$ is satisfied if and only if $(1+n)^{\frac{1}{\phi}} \leq \chi$ or $\phi \geq \bar{\phi}_1$.¹⁴ □

Appendix C. Proof of Proposition 3

The growth rate of per capita output is given by $g_{yt} = \frac{k_t[1+\mu(k_{t-1})]^{1+\phi}}{k_{t-1}(1+n)} - 1$.

Using Eqs. (B.1) and (B.2), we obtain $g_y^1 = (k^H)^{\frac{1-\alpha}{1+\alpha\phi}} (1+n)^{\frac{1-\alpha}{1+\alpha\phi}} - 1$ and $g_y^0 = (k^H)^{-\frac{1-\alpha}{1+\alpha\phi}} (1+n)^{-\frac{2\alpha}{1+\alpha\phi}} - 1$. It is possible to show that $g_y^1 > g_y^0$ holds if and only if $(1+n)^{\frac{1}{\phi}} < \alpha^{-\frac{\alpha}{1-\alpha}} - 1$. For $\alpha < 0.5$, $\alpha^{-\frac{\alpha}{1-\alpha}} - 1 < 1$ holds (Case(b)). Therefore, no positive pair of ϕ and n exists such that $(1+n)^{\frac{1}{\phi}} < \alpha^{-\frac{\alpha}{1-\alpha}} - 1$ is satisfied. On the other hand, when $\alpha > 1/2$, $\alpha^{-\frac{\alpha}{1-\alpha}} - 1$ is larger than 1 (Case(a)). In this case, $(1+n)^{\frac{1}{\phi}} < \alpha^{-\frac{\alpha}{1-\alpha}} - 1$ holds for $\phi > \bar{\phi}_2$. Since $\chi \geq \alpha^{-\frac{\alpha}{1-\alpha}} - 1$, $\bar{\phi}_2 \geq \bar{\phi}_1$ holds. □

Appendix D. Model with infinitely lived agents

In this appendix, we consider an infinitely-lived agent economy instead of an OLG framework, and show the robustness of our results. Using an OLG framework is unsuitable in the temporally patent model as in Matsuyama (1999) or Aloi and Lassele (2007), because the one period of the discrete time has two distinct interpretations:

patent length and half of a lifetime.¹⁵ There is no reason for these two interpretations to be identical.

D.1. The model

We assume the same structure as in Section 2, except for households. Therefore, the equations in Sections 2.1 through 2.3, Eqs. (2.2) and (2.3) hold, where L_t is the number of infinitely-lived households who each supply one unit of labor inelastically and grow at $n > 0$. The other parameters and variables are defined in Section 2.

As for consumers or households, assuming infinitely-lived agents implies that the optimal consumption path is characterized by an Euler equation, instead of the linear savings function. Each household chooses a consumption path that maximizes their discounted utility, $\sum_{t=0}^{\infty} \beta^t \log \tilde{c}_t$, subject to the budget constraint, $\tilde{k}_{t+1} = w_t + r_t \tilde{k}_t - \tilde{c}_t - n \tilde{k}_{t+1}$, where $\beta \in (0,1)$ is the discount factor. $\tilde{c}_t = C_t/L_t$ and $\tilde{k}_t = K_t/L_t$ are the per capita consumption and stock of capital, respectively. The final goods market clears when

$$Y_t = K_{t+1} + C_t. \quad (\text{D.1})$$

The solution to this simple maximization problem is characterized by an Euler equation and a transversality condition as follows:

$$\frac{\tilde{c}_t}{\tilde{c}_{t-1}} = \frac{\beta r_t}{1+n}, \quad (\text{D.2})$$

$$\lim_{T \rightarrow \infty} \beta^T \frac{\tilde{k}_{T+1}}{\tilde{c}_T} = 0. \quad (\text{D.3})$$

¹⁵ Matsuyama (1999) did not present utility maximization explicitly; however, his savings function can be derived from the conventional OLG assumptions. Matsuyama (2001) studied an infinitely-lived agent economy.

¹⁴ Since $\alpha^{\frac{\alpha}{1-\alpha}}(1-\alpha^{\frac{\alpha}{1-\alpha}}) \leq 0.25$, χ and $\bar{\phi}_1$ are real numbers.

D.2. Equilibrium

We define the new variables, $\hat{c}_{t-1} \equiv A^{\frac{1}{1+\phi}} \alpha^{\frac{\alpha}{1+\phi}} (1-\alpha) \eta_{t-1}^{\frac{1}{1+\phi}}$, and $c_{t-1} \equiv \alpha^{\frac{\alpha}{1+\phi}} (1-\alpha) \eta_{t-1}^{\frac{1}{1+\phi}}$. In equilibrium, $Y_t = r_t K_t + w_t L_t$ holds. Therefore the rate of return on capital is

$$r_t = \alpha \frac{Y_t}{K_t} = \alpha \hat{c}_{t-1}^{1-\alpha} \psi(k_{t-1}) k_{t-1}^{-1}. \quad (\text{D.4})$$

Summarizing Eqs. (2.2), (2.3), (D.1), (D.2), and (D.4) provides the three-dimensional dynamical system as follows:

$$\begin{aligned} k_t &= \hat{f}^k(k_{t-1}, \hat{c}_{t-1}, c_{t-1}) \equiv \begin{cases} \hat{c}_{t-1}^{1-\alpha} k_{t-1}^{\alpha} \left[1 - \alpha \beta \frac{c_{t-1}}{k_{t-1}} \right], & \text{for } k_{t-1} \leq 1, \\ \frac{\hat{c}_{t-1}^{1-\alpha} [k_{t-1} - \alpha \beta c_{t-1}]}{1 + \alpha \frac{1}{1-\alpha} (k_{t-1} - 1)}, & \text{for } k_{t-1} > 1, \end{cases} \\ \hat{c}_t &= \hat{f}^c(k_{t-1}, \hat{c}_{t-1}, c_{t-1}), \\ c_t &= f^c(k_{t-1}, \hat{c}_{t-1}, c_{t-1}) \equiv \begin{cases} \alpha \beta \hat{c}_{t-1}^{1-\alpha} k_{t-1}^{-(1-\alpha)} c_{t-1}, & \text{for } k_{t-1} \leq 1, \\ \frac{\alpha \beta \hat{c}_{t-1}^{1-\alpha} c_{t-1}}{1 + \alpha \frac{1}{1-\alpha} (k_{t-1} - 1)}, & \text{for } k_{t-1} > 1, \end{cases} \end{aligned} \quad (\text{D.5})$$

where $\hat{f}^c(\cdot, \cdot)$ was defined in Eq. (2.4). If the initial value of $(k_0, \hat{c}_0)$ is given, the equilibrium path $\{k_t, \hat{c}_t, c_t\}_0^\infty$ is characterized by the law of motion (Eq. (D.5)) and the transversality condition (Eq. (D.3)).

D.3. Dynamics

The law of motion (Eq. (D.5)) has a unique positive fixed point, $(k^*, \hat{c}^*, c^*)$, where

$$\hat{c}^* = \left[\frac{(1+n) \frac{1+\phi}{\alpha\beta}}{1-\alpha} \right]^{\frac{1}{1-\alpha}}, \quad c^* = k^* \left(\frac{1-\alpha\beta}{\alpha\beta} \right).$$

At this fixed point, the economy achieves balanced growth. Moreover, since $\hat{k}$ and $\hat{c}$ grow at the same rate along the BGP, the transversality condition (Eq. (D.3)) is satisfied as long as $\beta < 1$.

The three-dimensional system (Eq. (D.5)) has two predetermined variables, k and $\hat{c}$, and one non-predetermined variable, c . The local saddle path stability requires a two-dimensional locally stable manifold. We can verify the following proposition through a local stability analysis.

Proposition 4. *There is a unique threshold of ϕ , $\hat{\phi}_b$, which satisfies $\hat{B}(\hat{\phi}_b) - \Lambda(\hat{\phi}_b) = 0$. If $\phi > \hat{\phi}_b$, the fixed point $(k^*, \hat{c}^*, c^*)$ is locally unstable, where $\hat{B}(\phi)$ is defined as follows:*

$$\hat{B}(\phi) \equiv \frac{2(1+\alpha\beta) - \phi[\alpha^2\beta + \alpha(1+\beta) + 1]}{2(1+\alpha\beta) + \phi[\alpha^2\beta + \alpha(1+\beta) + 1]}.$$

According to the proposition, for a sufficiently large value of ϕ , only the one-dimensional locally stable manifold exists. Therefore, the economy that begins close to the fixed point will move away from it. Since a trajectory cannot approach the unique fixed point asymptotically, the equilibrium dynamics of the economy exhibits endogenous fluctuations for almost all initial conditions.

D.4. Period 2 cycles

With respect to the existence of the period 2 cycles, we can show a similar result with Proposition 2, i.e., the three-dimensional

dynamical system (Eq. (D.5)) has the period 2 cycles fluctuating between the R&D regime and the no R&D regime; this exists for $\phi > \hat{\phi}_1$. Such periodic orbits satisfy the transversality condition. As for the cyclical properties of R&D, it is possible to illustrate an identical result, as shown in Proposition 3 and Fig. 1. The discount factor β does not affect these results.

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